Dr.rer.nat. Jannik **Schestag**

Doctor rerum naturalium · Computer Science M.Sc. · Economical Mathematics B.Sc.

Postdoctoral researcher in theoretical computer science with a particular a focus on parameterized algorithms, algorithmic phylogenetics, and discrete mathematics. Experienced in international research collaboration, and interested in higher education teaching.

jannik.schestag@gmail.com | www.JannikSchestag.eu | Jannik Schestag

Academic Positions and Education

Post-Doctoral Research Position — *August 2026 - presumably July 2028*
Universitetet i Bergen — *Bergen, Norway*

Post-Doctoral Research Position — *January 2025 - June 2026*
Technische Universiteit Delft — *Delft, The Netherlands*

Ph.D. Computer Science — *May 2022 - January 2025*
Friedrich-Schiller-Universität Jena — *Jena, Germany*
Submitted: July 2024—Defended: January 2025—Final grade: *magna cum laude*

Visiting Researcher — *April 2023 - September 2023*
Technische Universiteit Delft — *Delft, The Netherlands*

M.Sc. Computer Science — *April 2019 - February 2022*
Philipps-Universität Marburg — *Marburg, Germany*
Final grade: 14,0 points (of 15); 1,0 grade (1,0 is the best)—Remark: very good

B.Sc. Computer Science — *April 2016 - December 2018*
Philipps-Universität Marburg — *Marburg, Germany*
Final grade: 13,4 points (of 15); 1,2 grade (1,0 is the best)—Remark: very good

B.Sc. Mathematical Economics — *October 2015 - June 2020*
Philipps-Universität Marburg — *Marburg, Germany*
Final grade: 11,5 points (of 15); 1,9 grade (1,0 is the best)—Remark: good

Research Funding and Grant Support

NWO Supported as a postdoctoral researcher by the Dutch Research Council (NWO), project OCENW.GROOT.2019.015 “Optimization for and with Machine Learning (OPTIMAL)”, 2025 and 2026.

DAAD *Principal applicant*. Awarded competitive full funding by the German Academic Exchange Service (DAAD), project no. 57556279, for a six-month international research visit at TU Delft, 2023.

Program Committee Member

2027 The 52nd International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM 2027)

Reviewed Articles for

2026 The 20th Scandinavian Symposium on Algorithm Theory (SWAT 2026)
2026 The 52nd International Workshop on Graph-Theoretic Concepts in Computer Science (WG 2026); two articles
2026 The 24th Symposium on Experimental Algorithms (SEA 2026)
2025 The 24th International Workshop on Algorithms in Bioinformatics (WABI 2025)
2024 The 23rd International Workshop on Algorithms in Bioinformatics (WABI 2024)
2024 Journal of Computer and System Sciences
2023 Discrete Mathematics & Theoretical Computer Science

Journal Publications

[1] **A multivariate complexity analysis of the Generalized Noah's Ark Problem**
Discrete Applied Mathematics (vol. 382) *2025*
Christian Komusiewicz and *Jannik Schestag*
DOI: 10.1016/j.dam.2025.11.037; ArXiv: 2307.03518; Journal version of [16].

[2] **On Critical Node Problems with Vulnerable Vertices** (My Master's Thesis)
Journal of Graph Algorithms and Applications (vol. 28 no. 1) *2024*
Niels Grüttemeier, Christian Komusiewicz, *Jannik Schestag*, and Frank Sommer
DOI: 10.7155/jgaa.v28i1.2922; Journal version of [17].

[3] **Destroying Bicolored P_3s by Deleting Few Edges** (My Bachelor's Thesis)
Discrete Mathematics & Theoretical Computer Science (vol. 23 no. 1) *2021*
Niels Grüttemeier, Christian Komusiewicz, *Jannik Schestag*, and Frank Sommer
DOI: 10.46298/dmtcs.6108; ArXiv: 1901.03627; Journal version of [18].

Conference Publications

[4] **PaNDA: Efficient Optimization of Phylogenetic Diversity in Networks**
[*Paper accepted for presentation; RECOMB did not publish formal proceedings in 2026*]
The 30th Annual International Conference on Research in Computational Molecular Biology (RECOMB 2026), Thessaloniki, Greece, May 2026. *2026*
Niels Holtgrefe, Leo van Iersel, Ruben Meuwese, Yukihiro Murakami, and *Jannik Schestag*
DOI: 10.1101/2025.11.14.688467

[5] **Limits of Kernelization and Parametrization for Phylogenetic Diversity with Dependencies**
[*Accepted, publication forthcoming*]
In *Proceedings of the 17th Latin American Theoretical Informatics Symposium (LATIN 2026)*, Florianópolis, Brazil, April 2026. LNCS, Springer. *2026*
Niels Holtgrefe, *Jannik Schestag*, and Norbert Zeh
ArXiv: 2602.12959

[6] **Weighted Food Webs Make Computing Phylogenetic Diversity So Much Harder**
In *Proceedings of the 51st International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM 2026)*, Kraków, Poland, February 2026. LNCS, Springer. *2026*
Jannik Schestag
DOI: 10.1007/978-3-032-17801-5_14; ArXiv: 2510.05911

[7] **Parameterized Algorithms for Diversity of Networks with Ecological Dependencies**
In *Proceedings of the 20th International Symposium on Parameterized and Exact Computation (IPEC 2025)*, Warsaw, Poland, September 2025. LIPIcs, Dagstuhl. *2025*
Mark Jones and *Jannik Schestag*
DOI: 10.4230/LIPIcs.IPEC.2025.11; ArXiv: 2510.09512

[8] **Average-Tree Phylogenetic Diversity of Networks**
In *Proceedings of the 24th International Workshop on Algorithms in Bioinformatics (WABI 2025)*, College Park, Maryland, USA, August 2025. LIPIcs, Dagstuhl. *2025*
Leo van Iersel, Mark Jones, *Jannik Schestag*, Celine Scornavacca, and Mathias Weller
DOI: 10.4230/LIPIcs.WABI.2025.15; HAL Id: hal-05194249

[9] **Phylogenetic Network Diversity Parameterized by Reticulation Number and Beyond**
In *Proceedings of the 23rd RECOMB International Workshop on Comparative Genomics (RECOMB-CG 2025)*, Seoul, Korea Republic, April 2025. LNBI, Springer. *2025*
Leo van Iersel, Mark Jones, *Jannik Schestag*, Celine Scornavacca, and Mathias Weller
DOI: 10.1007/978-3-031-94928-9_7; ArXiv: 2405.01091

[10] **Maximizing Phylogenetic Diversity under Ecological Constraints: A Parameterized Complexity Study**

In *Proceedings of the 44th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2024)*, Gandhinagar, India, December 2024. LIPIcs, Dagstuhl. *2024*

Christian Komusiewicz and *Jannik Schestag*

DOI: 10.4230/LIPIcs.FSTTCS.2024.28; ArXiv: 2405.17314

[11] **Protective and Nonprotective Subset Sum Games: A Parameterized Complexity Analysis**

In *Proceedings of the 8th International Conference on Algorithmic Decision Theory (ADT 2024)*, Piscataway, New Jersey, United States, October 2024. LNCS, Springer. *2024*

Jaroslav Garvardt, Ber Lorke, Christian Komusiewicz, and *Jannik Schestag*

DOI: 10.1007/978-3-031-73903-3_6

[12] **On the Complexity of Finding a Sparse Connected Spanning Subgraph in a Non-Uniform Failure Model**

In *Proceedings of the 18th International Symposium on Parameterized and Exact Computation (IPEC 2023)*, Amsterdam, The Netherlands, September 2023. LIPIcs, Dagstuhl. *2023*

Matthias Bentert, *Jannik Schestag*, and Frank Sommer

DOI: 10.4230/LIPIcs.IPEC.2023.4; ArXiv: 2308.04575; Conference version of [19].

[13] **How Can We Maximize Phylogenetic Diversity? Parameterized Approaches for Networks**

In *Proceedings of the 18th International Symposium on Parameterized and Exact Computation (IPEC 2023)*, Amsterdam, The Netherlands, September 2023. LIPIcs, Dagstuhl. *2023*

Mark Jones and *Jannik Schestag*

DOI: 10.4230/LIPIcs.IPEC.2023.30; TU Delft repository

[14] **Finding Degree-Constrained Acyclic Orientations**

In *Proceedings of the 18th International Symposium on Parameterized and Exact Computation (IPEC 2023)*, Amsterdam, The Netherlands, September 2023. LIPIcs, Dagstuhl. *2023*

Jaroslav Garvardt, Malte Renken, *Jannik Schestag*, and Mathias Weller

DOI: 10.4230/LIPIcs.IPEC.2023.19; TU Delft repository

[15] **On the Complexity of Parameterized Local Search for the Maximum Parsimony Problem**

In *Proceedings of the 34th Annual Symposium on Combinatorial Pattern Matching (CPM 2023)*, Marne-la-Vallée, France, June 2023. LIPIcs, Dagstuhl. *2023*

Christian Komusiewicz, Simone Linz, Nils Morawietz, and *Jannik Schestag*

DOI: 10.4230/LIPIcs.CPM.2023.18

[16] **A Multivariate Complexity Analysis of the Generalized Noah's Ark Problem**

In *Proceedings of the 19th Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW 2023)*, Garmisch-Partenkirchen, Germany, June 2023. LNCS, Springer. *2023*

Christian Komusiewicz and *Jannik Schestag*

DOI: 10.1007/978-3-031-46826-1_9; ArXiv: 2307.03518; Conference version of [1].

[17] **On Critical Node Problems with Vulnerable Vertices** (My Master's Thesis)

In *Proceedings of the 33rd International Workshop on Combinatorial Algorithms (IWOCA 2022)*, Trier, Germany, June 2022. LNCS, Springer. *2022*

Niels Grüttemeier, Christian Komusiewicz, *Jannik Schestag*, and Frank Sommer

DOI: 10.1007/978-3-031-06678-8_36; Conference version of [2].

[18] **Destroying Bicolored P_3s by Deleting Few Edges** (My Bachelor's Thesis)

In *Proceedings of the 15th Conference on Computability in Europe (CiE 2019)*, Durham, United Kingdom, July 2019. LNCS, Springer. *2019*

Niels Grüttemeier, Christian Komusiewicz, *Jannik Schestag*, and Frank Sommer

DOI: 10.1007/978-3-030-22996-2_17; ArXiv: 1901.03627; Conference version of [3].

Submitted Journal Articles

[19] **Finding a Sparse Connected Spanning Subgraph in a non-Uniform Failure Model**
Algorithmica
Matthias Bentert, *Jannik Schestag*, and Frank Sommer
ArXiv: 2308.04575; Journal version of [12].

[20] **Maximizing Phylogenetic Diversity under Time Pressure: Planning with Imminent Extinctions**
Journal of Combinatorial Optimization
Mark Jones and *Jannik Schestag*
ArXiv: 2403.14217; A preliminary version was presented at ECCO '25 which had no precedings.

Research Interests

Fixed-Parameter Tractability
Algorithmic Lower Bounds
Phylogenetics
Number-Theoretic Problems

Administrative Information

Citizenship	EU Citizen and Passport Holder
Languages	German (Native), English (Fluent), Dutch (Proficient), Norwegian (Intermediate), Swedish (Intermediate), French (Intermediate)
Teaching Languages	English, German, and Dutch

Held Scientific Conference Talks

Orienting Unrooted Phylogenetic Networks — *July 2026*
Uniting Phylogenetic Network Research — *Lorentz Center, Leiden, The Netherlands*

Limits of Kernelization and Parametrization for Phylogenetic Diversity with Dependencies — *April 2026*
The 17th Latin American Theoretical Informatics Symposium (LATIN 2026) — *Florianópolis, Brazil*
Presentation of [5].

Generalizing Phylogenetic Diversity to Networks — *March 2026*
International Workshop on Probabilistic, Combinatorial, and Algorithmic Phylogenetics — *Taipei, Taiwan*

Weighted Food Webs Make Computing Phylogenetic Diversity So Much Harder — *February 2026*
The 51st International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM 2026) — *Kraków, Poland*
Presentation of [6].

Parameterized Algorithms for Diversity of Networks with Ecological Dependencies — *September 2025*
The 20th International Symposium on Parameterized and Exact Computation (IPEC 2025) — *Warsaw, Poland*
Presentation of [7].

Average-Tree Phylogenetic Diversity of Networks — *August 2025*
The 25th International Workshop on Algorithms in Bioinformatics (WABI 2025) — *College Park, MD, United States*
Presentation of [8].

Maximizing Phylogenetic Diversity under Time Pressure: Planning with Imminent Extinctions — *May 2025*
The 38th Conference of the European Chapter of Combinatorial Optimization (ECCO 2025) — *Marrakesh, Morocco*
Presentation of [20].

Phylogenetic Network Diversity Parameterized by Reticulation Number and Beyond *April 2025*

The 23rd RECOMB International Workshop on Comparative Genomics (RECOMB-CG 2025) *Seoul, Korea Republic*

Presentation of [9].

Maximizing Phylogenetic Diversity under Ecological Constraints: A Parameterized Complexity Study *December 2024*

The 44th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2024) *Gandhinagar, India*

Presentation of [10].

How can we Maximize Phylogenetic Diversity? Parameterized Approaches for Networks *September 2023*

The 18th International Symposium on Parameterized and Exact Computation (IPEC 2023) *Amsterdam, The Netherlands*

Presentation of [13].

Finding Degree-Constrained Acyclic Orientations *September 2023*

The 18th International Symposium on Parameterized and Exact Computation (IPEC 2023) *Amsterdam, The Netherlands*

Presentation of [14].

A Multivariate Complexity Analysis of the Generalized Noah's Ark Problem *June 2023*

The 19th Cologne-Twente Workshop on Graphs and Combinatorial Optimization (CTW 2023) *Garmisch-Partenkirchen, Germany*

Presentation of [16,1].

On Critical Node Problems with Vulnerable Vertices *June 2022*

The 33rd International Workshop on Combinatorial Algorithms (IWOCA 2022) *Trier, Germany*

Presentation of [17,2].